Experiment 16

Title

Write a Prolog Program to Sum the Integers from 1 to n.

Aim

To develop a Python program to implement a Feed Forward Neural Network for classifying the Iris dataset.

Objective

To train a neural network using the Iris dataset and evaluate its classification accuracy.

Algorithm

Step 1: Import the required libraries.

Step 2: Load the Iris dataset.

Step 3: Split the dataset into training and testing sets.

Step 4: Create a Feed Forward Neural Network with one hidden layer and one output layer.

Step 5: Compile the model using the Adam optimizer and categorical crossentropy loss function.

Step 6: Train the model using the training dataset.

Step 7: Test the model using the testing dataset.

Step 8: Display the classification accuracy.

Program

```
from tensorflow.keras.models import Sequential

from tensorflow.keras.layers import Dense

from tensorflow.keras.utils import to_categorical

from sklearn.datasets import load_iris

from sklearn.model_selection import train_test_split


# Load Iris Dataset
```

```
iris = load_iris()

X = iris.data
y = iris.target

# Convert target values into one-hot encoding
y = to_categorical(y)

# Split the dataset
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42
)

# Create Feed Forward Neural Network
model = Sequential()

# Hidden Layer
model.add(Dense(10, input_shape=(4,), activation='relu'))

# Output Layer
model.add(Dense(3, activation='softmax'))

# Compile Model
model.compile(
    optimizer='adam',
    loss='categorical_crossentropy',
    metrics=['accuracy']
)
```

```
# Train Model
model.fit(
    X_train,
    y_train,
    epochs=50,
    batch_size=10,
    verbose=1
)

# Evaluate Model
loss, accuracy = model.evaluate(X_test, y_test, verbose=0)

print("\nModel Evaluation")
print("----------------")
print("Testing Accuracy : {:.2f}%".format(accuracy * 100))
print("Testing Loss     : {:.4f}".format(loss))
```

Sample Input

No manual input is required.

The Iris dataset is loaded automatically by the program.

Sample Output

```
[1m 1/11[0m [32m   ━[0m[37m ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m [1m0s[0m 3
3ms/step - accuracy: 0.9000 - loss: 0.5421[1m11/11[0m [32m
━━━━━━━━━━━━━━━━━━━━[0m[37m[0m [1m0s[0m 5ms/step - accuracy: 0.9238 - loss: 0.4515
Epoch 45/50
[1m 1/11[0m [32m   ━[0m[37m ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m [1m0s[0m 3
4ms/step - accuracy: 1.0000 - loss: 0.3697[1m11/11[0m [32m
━━━━━━━━━━━━━━━━━━━━[0m[37m[0m [1m0s[0m 3ms/step - accuracy: 0.9333 - loss: 0.4478
Epoch 46/50
[1m 1/11[0m [32m   ━[0m[37m ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m [1m0s[0m 4
3ms/step - accuracy: 1.0000 - loss: 0.3865[1m11/11[0m [32m
━━━━━━━━━━━━━━━━━━━━[0m[37m[0m [1m0s[0m 4ms/step - accuracy: 0.9238 - loss: 0.4430
Epoch 47/50
[1m 1/11[0m [32m   ━[0m[37m ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m [1m0s[0m 4
0ms/step - accuracy: 1.0000 - loss: 0.3683[1m11/11[0m [32m
━━━━━━━━━━━━━━━━━━━━[0m[37m[0m [1m0s[0m 5ms/step - accuracy: 0.9143 - loss: 0.4383
Epoch 48/50
[1m 1/11[0m [32m   ━[0m[37m ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m [1m0s[0m 3
4ms/step - accuracy: 0.9000 - loss: 0.5365[1m11/11[0m [32m
━━━━━━━━━━━━━━━━━━━━[0m[37m[0m [1m0s[0m 6ms/step - accuracy: 0.9333 - loss: 0.4362
Epoch 49/50
[1m 1/11[0m [32m   ━[0m[37m ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m [1m0s[0m 3
9ms/step - accuracy: 0.9000 - loss: 0.4972[1m11/11[0m [32m
━━━━━━━━━━━━━━━━━━━━[0m[37m[0m [1m0s[0m 4ms/step - accuracy: 0.9524 - loss: 0.4296
Epoch 50/50
[1m 1/11[0m [32m   ━[0m[37m ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m [1m1s[0m 1
11ms/step - accuracy: 0.9000 - loss: 0.4240[1m11/11[0m [32m
━━━━━━━━━━━━━━━━━━━━[0m[37m[0m [1m0s[0m 6ms/step - accuracy: 0.9143 - loss: 0.4272

Model Evaluation
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Testing Accuracy : 93.33%
Testing Loss     : 0.3586
```

Result

Thus, the Python program to implement a **Feed Forward Neural Network** was executed successfully, and the classification accuracy of the Iris dataset was obtained.